

Supplementary material: Path-spread-resolved quantum transition-state theory for proton transfer

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1 Proof of the harmonic continuum formula

The Euclidean action of a harmonic oscillator $V = \frac{1}{2}m\omega^2 q^2$ is diagonal in Matsubara modes

$$q(\tau) = \frac{1}{\beta\hbar} \sum_{n=-\infty}^{\infty} \tilde{q}_n e^{i\omega_n \tau}, \quad \omega_n = \frac{2\pi n}{\beta\hbar}. \quad (1)$$

Each mode is Gaussian with $\langle |\tilde{q}_n|^2 \rangle = k_B T \beta\hbar / [m(\omega_n^2 + \omega^2)]$. The equal-time correlator is

$$G(0) = \frac{k_B T}{m} \sum_{n=-\infty}^{\infty} \frac{1}{\omega_n^2 + \omega^2} = \frac{\hbar}{2m\omega} \coth\left(\frac{\beta\hbar\omega}{2}\right), \quad (2)$$

where the sum is the standard cotangent partial-fraction identity. The $n = 0$ sector is the centroid, with variance $k_B T / (m\omega^2)$, which is the classical Boltzmann result for a harmonic well.^[1] Subtracting gives

$$\langle \chi_p \rangle = G(0) - \langle \xi_p^2 \rangle, \quad (3)$$

which is Eq. (8) of the main text. Non-negativity follows from $\coth x \geq 1/x$.

Low T : $\coth(\beta\hbar\omega/2) \rightarrow 1$ and the centroid term vanishes, so $\langle \chi_p \rangle \rightarrow \hbar/(2m\omega)$. High T : $\coth x = 1/x + x/3 + O(x^3)$ yields $\langle \chi_p \rangle \rightarrow \hbar^2\beta/(12m) + O(\beta^3)$.

2 Exact finite-bead formula

The P -bead ring polymer is a cyclic Gaussian with eigenvalues

$$\lambda_k = \frac{4mP}{\hbar^2\beta^2} \sin^2(\pi k/P) + \frac{m\omega^2}{P}, \quad k = 0, \dots, P-1. \quad (4)$$

The $k = 0$ eigenvalue is $m\omega^2/P$ and produces the centroid variance $k_B T / (m\omega^2)$. The primitive bead variance minus the centroid variance is

$$\langle \chi_p \rangle_P = \frac{1}{P} \sum_{k=1}^{P-1} \frac{1}{\beta\lambda_k}. \quad (5)$$

As $P \rightarrow \infty$ the sum becomes the Matsubara sum of the previous section.

3 Isotope ordering (harmonic)

With $k = m\omega^2$ fixed, $\omega(m) = \sqrt{k/m}$ and

$$g(m) = \frac{\hbar}{2\sqrt{km}} \coth\left(\frac{\beta\hbar\sqrt{k/m}}{2}\right) - \frac{k_B T}{k}. \quad (6)$$

The second term is mass-independent. The first is $(k_B T/k) u \coth u$ with $u = \beta\hbar\sqrt{k/m}/2$. Since $u \coth u$ is strictly increasing for $u > 0$ and u is strictly decreasing in m , $g(m)$ is strictly decreasing, so $g(m_H) > g(m_D)$. The $T \rightarrow 0$ and $T \rightarrow \infty$ ratios follow by substituting the limits of the previous section. These statements are harmonic; they are not theorems for a general anharmonic surface.

4 Computational parameters

Table 1: Production parameters.

Quantity	Value
Harmonic ω	1000 cm ⁻¹
Harmonic P (H, D)	64, 32
Harmonic production steps	8×10^4
Eckart width a	0.50 Å
Eckart T	300 K
Eckart PIMC P (H, D)	32, 24
Eckart PIMC umbrellas	11 windows in $[-2, 2]$ Å
Eckart PIMC k_{umb}	60 kJ mol ⁻¹ Å ⁻²
Eckart PIMC steps per window	4×10^4 (8×10^3 equilibration)
ADW q_0	0.70 Å
ADW α	0.025 kJ mol ⁻¹ Å ⁻¹
ADW umbrellas	11 windows in $[-1, 1]$ Å
ADW k_{umb}	80 kJ mol ⁻¹ Å ⁻²
ADW P (H, D)	32, 24
ADW steps per window	5×10^4
RNG seed	42

Staging proposals use the sequential Lévy construction of the free-particle bridge (end-points of a segment of length $L = P/4$ held fixed; interior beads drawn conditionally).

Independent sampling of Brownian-bridge marginals is incorrect and inflates χ_p at large P .

5 Eckart centroid QTST

On the symmetric Eckart barrier the centroid QTST rate is

$$k_{\text{cTST}} = \sqrt{k_{\text{B}}T/(2\pi m)} e^{-\beta F(0)},$$

with F the WHAM potential of mean force zeroed on the $|\bar{\xi}_p| > 1.6 \text{ \AA}$ tails. With a class-independent transmission coefficient $\kappa = 1$, the QRR integral of the main text is identically k_{cTST} . Residual tail scatter in F is of order 0.5 kJ mol^{-1} . Deuterium at $V_0 = 12 k_{\text{B}}T$ is excluded from the rate comparison: the dividing-surface window drifted and produced $F_D(0) < F_H(0)$, which is unphysical. Path-class fractions f_{deloc} at that point remain consistent with the other deuterium barriers and are retained in Fig. 4(c) of the main text.

The isolated asymmetric double well is bound. It has no irreversible thermal rate, so DVR “rates” on that potential are not a benchmark.

6 Reproducibility

```
python run_all.py --n_steps 80000 --n_equil 15000
python run_all.py --skip_experiments    # figures + compile
```

Dependencies: `numpy`>=1.26, `scipy`>=1.12, `matplotlib`>=3.8. All JSON result files used to generate the figures are in `results/`.

References

- [1] J. Cao and G. A. Voth. The formulation of quantum statistical mechanics based on the Feynman path centroid density. I. Equilibrium properties. *J. Chem. Phys.*, 100:5106–5117, 1994.